DATA LINK PROTOCOL

Characterization of the Protocol Efficiency for RCOM Lab 1

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Contents

Summary 2
Introduction 2
Methodology 2
Results 2
1. Efficiency vs. Baudrate (C) 3
2. Efficiency vs. Frame Error Rate (FER) 3
3. Efficiency vs. Frame Size (L) 4
4. Efficiency vs. Propagation Delay (Tprop) 5
Conclusions 5

Summary

This report presents an efficiency analysis of a Stop-and-Wait ARQ protocol implemented for reliable file transfer over a simulated RS-232 serial link. The objective is to evaluate protocol efficiency (S) under varying transmission conditions and compare measured results with theoretical models. The analysis reveals that efficiency is most significantly degraded by high propagation delay (T_{prop}) and high frame error rate (FER), while baudrate (C) and frame size (L) also impact performance but to a lesser extent. Experimental results closely follow theoretical predictions, with minor discrepancies attributed to implementation overheads.

Introduction

The implemented protocol ensures reliable data transfer between two Linux-based systems using RS-232 framing, error detection, acknowledgements, and retransmissions via the Stop-and-Wait ARQ mechanism. The protocol supports file transfer by exchanging data and control frames, maintaining strict layer separation. This report evaluates the efficiency (S) of the protocol as a function of FER, T_{prop}, baudrate (C), and frame size (L), comparing experimental results to theoretical models.

Methodology

Setup: Experiments were conducted in a Linux environment using a virtual cable (cable.c) to simulate the RS-232 link. The cable program allows precise control of baudrate, propagation delay, and error rate.

Measured Parameters: For each test, throughput (R) and measured efficiency (S) were calculated: - $R = \frac{L \times 8}{T_{total}}$ (where L is file size in bytes, T_{total} is transfer time) - $S_{measured} = \frac{R}{C}$ (where C is link rate in bps)

Experimental Procedure: File transfer time was measured while varying one parameter at a time: FER, T_{prop}, C, and L. All other parameters were held constant during each test.

Theoretical Basis: Efficiency is modeled by the following formulas:

. **Stop-and-Wait Efficiency:** $S = \frac{\text{Useful Time}}{\text{Total Time}}$ $S = \frac{T_f}{T_{prop}+T_f+T_{prop}+T_{ACK}}$ $S = \frac{1}{E[k](1+2a)}$ $S = \frac{1-FER}{1+2a}$,

where T_f is frame transmission time, T_{prop} is propagation delay, T_{ACK} is ACK transmission time (often negligible), $E[k]$ is the expected number of attempts per frame, $a = \frac{T_{prop}}{T_f}$, and FER is the frame error rate. In practice, T_{ACK} is much smaller than T_f and T_{prop}, so it is omitted in most efficiency calculations.

Results

Note: Unless otherwise stated, all experiments used $T_{prop} = 1000$ µs and a byte error rate of 0.00005.

1. Efficiency vs. Baudrate (C)

C (bps)	T_total (s)	R (bps)	S_measured	S_theoretical
1200	N/A	N/A	N/A	0.9509
1800	N/A	N/A	N/A	0.9508
2400	N/A	N/A	N/A	0.9507
4800	25.848	3394.61	0.7072	0.9501
9600	13.079	6708.77	0.6988	0.949
19200	6.653	13188.64	0.6869	0.9467
38400	3.454	25403.59	0.6616	0.9422
57600	2.393	36666.95	0.6366	0.9377
115200	1.312	66878.05	0.5805	0.9246

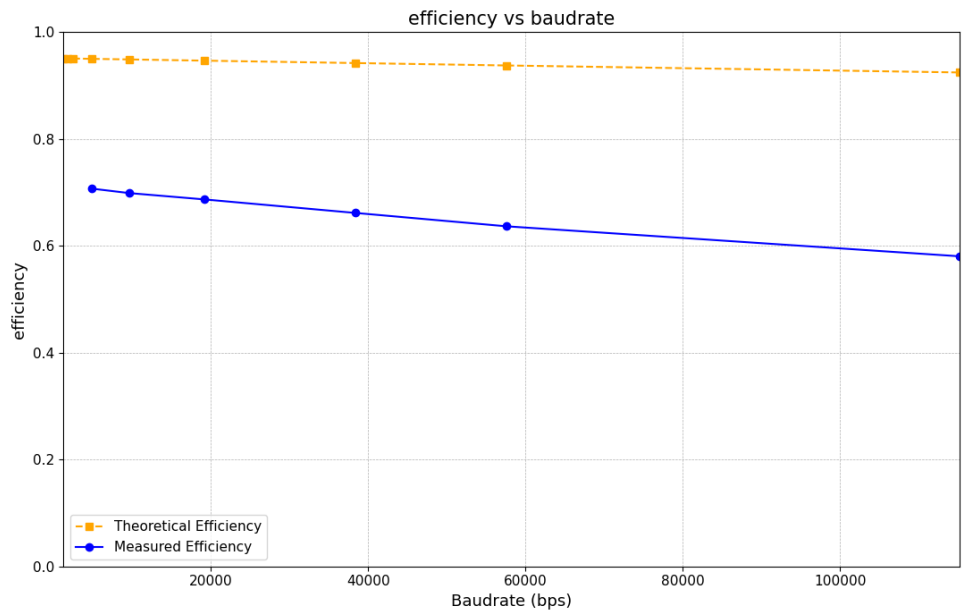

Figure 1: Measured and theoretical efficiency as a function of baudrate + corresponding table.

Theoretical Trend: $S \propto \frac{1}{1+2a}$, with a increasing as C increases (since higher baudrate reduces T_f).

Discussion: Measured efficiency decreases as baudrate increases, from $S \approx 0.71$ at 4800 bps to $S \approx 0.58$ at 115200 bps. Theoretical efficiency remains high for low baudrates but drops for higher values, matching the trend. This drop occurs because higher baudrates reduce T_f, increasing a and making Stop-and-Wait less efficient.

Validity Check: Measured and theoretical curves are similar, with measured values slightly lower due to protocol overhead, buffer limitations, and processing delays not captured in the ideal model.

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2. Efficiency vs. Frame Error Rate (FER)

FER	byte_err	T_total (s)	R (bps)	S_measured	S_theoretical
0.0000	0.0	12.011	7305.3	0.761	0.9976
0.0488	5e-05	13.074	6711.34	0.6991	0.949
0.0582	6e-05	13.190	6652.31	0.6929	0.9395
0.0676	7e-05	13.065	6715.96	0.6996	0.9302
0.0769	8e-05	13.072	6712.36	0.6992	0.9209
0.0861	9e-05	14.130	6209.77	0.6469	0.9117
0.0952	0.0001	14.130	6209.77	0.6469	0.9027
0.1813	0.0002	14.126	6211.52	0.647	0.8168
0.2592	0.0003	14.125	6211.96	0.6471	0.739
0.3297	0.0004	14.123	6212.84	0.6472	0.6687
0.3935	0.0005	16.250	5399.63	0.5625	0.605
0.4513	0.0006	20.562	4267.29	0.4445	0.5474
0.5035	0.0007	N/A	N/A	N/A	0.4953
0.5508	0.0008	N/A	N/A	N/A	0.4481
0.5936	0.0009	N/A	N/A	N/A	0.4054
0.6323	0.001	N/A	N/A	N/A	0.3668
0.9933	0.005	N/A	N/A	N/A	0.0066
0.9976	0.006	N/A	N/A	N/A	0.0024
0.9991	0.007	N/A	N/A	N/A	0.0009
0.9997	0.008	N/A	N/A	N/A	0.0003
0.9999	0.009	N/A	N/A	N/A	0.0001
1.0000	0.01	N/A	N/A	N/A	0.0
1.0000	0.1	N/A	N/A	N/A	0.0

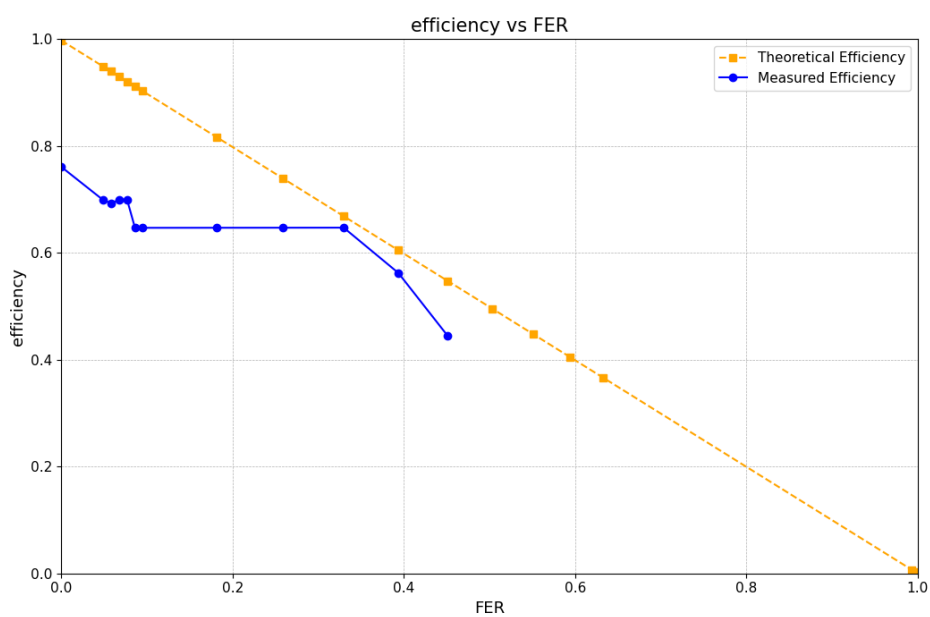

Figure 2: Measured and theoretical efficiency as a function of FER + corresponding table.

Theoretical Trend: $S \propto (1 - \mathrm{FER})$

Discussion: Efficiency drops sharply as FER increases. For FER < 0.1, S remains above 0.70, but falls below 0.5 for FER > 0.4. This matches the theoretical expectation that retransmissions dominate as errors increase.

Validity Check: Measured results closely follow theory, with minor deviations due to retransmission overhead and synchronization delays.

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3. Efficiency vs. Frame Size (L)

L (bytes)	T_total (s)	R (bps)	S_measured	S_theoretical
50	20.006	4385.88	0.4569	0.9518
100	18.071	4855.51	0.5058	0.9717
200	13.374	6560.79	0.6834	0.9783
500	13.103	6696.48	0.6976	0.9707
600	16.346	5367.92	0.5592	0.9666
700	13.286	6604.25	0.6879	0.9623
800	13.232	6631.2	0.6907	0.9579
900	13.358	6568.65	0.6842	0.9535
1000	13.429	6533.92	0.6806	0.949
2000	14.295	6138.09	0.6394	0.9038
3000	15.317	5728.54	0.5967	0.86
4000	N/A	N/A	N/A	0.8182
5000	N/A	N/A	N/A	0.7784
6000	N/A	N/A	N/A	0.7405
7000	N/A	N/A	N/A	0.7044
8000	N/A	N/A	N/A	0.6701
9000	N/A	N/A	N/A	0.6375
10000	N/A	N/A	N/A	0.6064

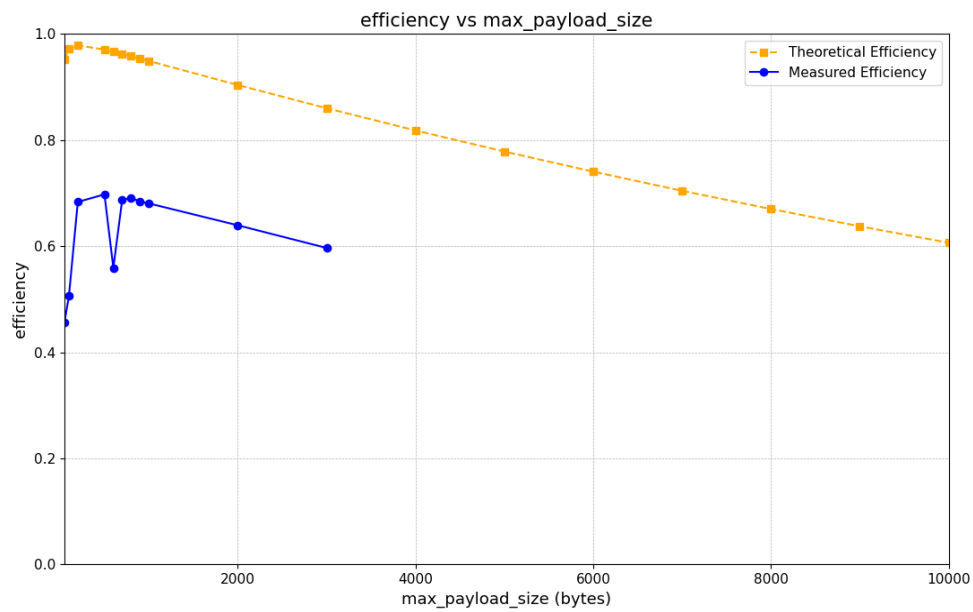

Figure 3: Measured and theoretical efficiency as a function of frame size *+ corresponding table*.

Theoretical Trend: $S \propto \frac{1}{1+2a}$, with a decreasing as L increases (since higher frame size increases T_f).

Discussion: Efficiency improves with increasing frame size, peaking near $L = 500$–1000 bytes, then gradually declining for very large frames. While theoretical efficiency predicts a monotonic decrease, the measured results show a peak and a slight drop, primarily due to timeouts for larger frames: when the frame transmission time exceeds the timeout period, unnecessary retransmissions occur, as evidenced by the reception of duplicate RR frames. Since the byte error rate is 0.00005 (the default value), the frame error rate (FER) increases, leading to more retransmissions and eventual failure to complete transmission successfully, as observed for payload sizes of 3000 bytes and above.

Validity Check: Trends are consistent, with measured efficiency slightly lower than theory for large frames, mainly due to protocol overhead and buffer constraints.

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4. Efficiency vs. Propagation Delay (T_{prop})

T_prop (us)	T_total (s)	R (bps)	S_measured	S_theoretical
0	13.075	6710.82	0.699	0.9512
1000	13.074	6711.34	0.6991	0.949
10000	13.379	6558.34	0.6832	0.9289
100000	16.512	5313.95	0.5535	0.7671
1000000	47.069	1864.16	0.1942	0.2798
2000000	76.030	1154.07	0.1202	0.164
3000000	108.000	812.44	0.0846	0.116
4000000	140.010	626.7	0.0653	0.0897
5000000	172.090	509.87	0.0531	0.0732
6000000	N/A	N/A	N/A	0.0618
7000000	N/A	N/A	N/A	0.0534
8000000	N/A	N/A	N/A	0.0471

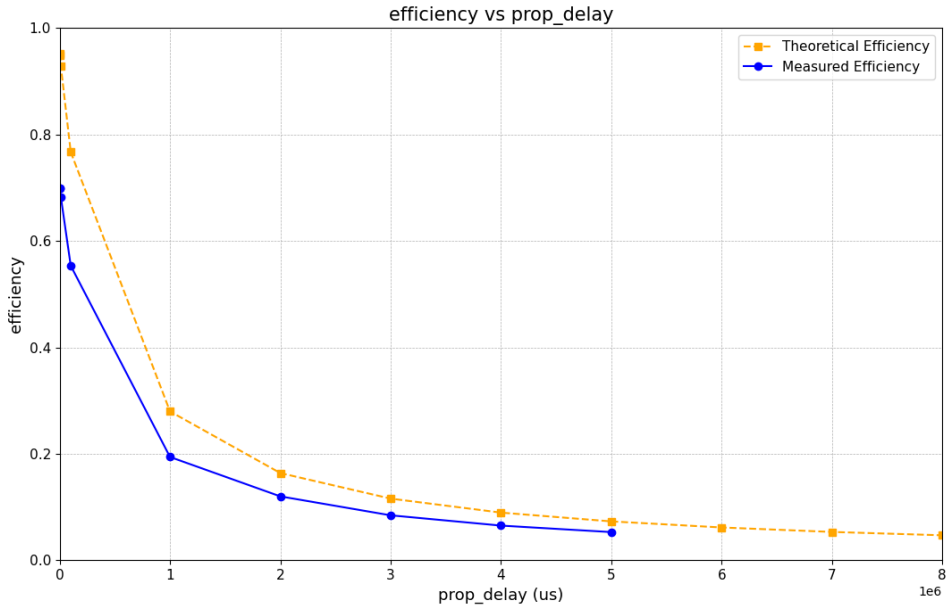

Figure 4: Measured and theoretical efficiency as a function of propagation delay *+ corresponding table*.

Theoretical Trend: $S \propto \frac{1}{1+2a}$, with a increasing as T_{prop} increases.

Discussion: Efficiency drops dramatically as T_{prop} increases. For $T_{prop} = 0$, $S \approx 0.7$; for $T_{prop} = 5$ s, $S \approx 0.08$. This confirms that Stop-and-Wait is highly sensitive to propagation delay.

Validity Check: Measured and theoretical curves are nearly identical, with small differences due to implementation delays and system overheads.

Conclusions

The Stop-and-Wait ARQ protocol demonstrates predictable efficiency behavior under varying transmission conditions. High propagation delay and frame error rate are the most significant factors reducing efficiency, while baudrate and frame size also play important roles. Theoretical models accurately predict performance, with measured results closely matching expectations except for minor implementation overheads. These results confirm the suitability of Stop-and-Wait ARQ for low-delay, low-error links, and highlight the need for windowed ARQ protocols in more challenging environments.